

MODULE -2
THEORY OF COST
COST & COST FUNCTION

Cost is the payments made to the factors of production which are used in the production of that commodity.

Cost function shows the functional relationship between output and cost of production. It gives the least cost combination of inputs corresponding to different levels of output. Cost function is given as:

$$C=f(Q)$$

Where C= Cost and Q=output

Differentiate Short & Long run cost curves. (6)

SHORT-RUN COST & LONG- RUN COSTS

The time factor is very important in the theory of cost. The time period divided cost into two categories short- run and long- run

<u>SHORT-RUN COSTS</u>	<u>LONG-RUN COSTS</u>
The short run costs are the costs over a period during which some of the factors are fixed in supply, like plant, machinery etc. It is a sum total of fixed cost and variable cost incurred by the producer in producing a commodity.	The long run costs are the costs over a period long enough to permit changes in all factors of production . In this period the firm can increase production by using more of all factors. Supply of a commodity can adjusted to changes in demand.

EXPLICIT COST & IMPLICIT COST

Explicit cost or direct cost is the actual **expenditure incurred by a firm to purchase or hire the inputs it needs in the production process. These inputs do not belong to the firm itself.** These include wages, rent interest payments etc.

Implicit Cost or imputed cost **is the cost of inputs owned by the firm** and used by the firm in its own production process. Besides purchasing or hiring resources from others, a producer may also use his own factor services in the process of production. For example the producer may contribute his own land, and may provide his own entrepreneurial and managerial services.

<u>IMPLICIT COST</u>	<u>EXPLICIT COST</u>
Implicit Cost is the opportunity cost, which is incurred when the entity uses the owner's resources like capital inventory etc.	Explicit Cost is incurred when the entity has to pay for the utilization of factors of production.
It helps in the calculation of only economic profit.	It helps in the calculation of both economic and accounting profit.
It is neither recorded nor reported to the management of the company.	Explicit Cost is recorded and reported to the management.
Examples :Wages of self-labor, rent for self-owned premises etc.	Example: Wages ,rent ,interest on loan etc.

Differentiate fixed cost & variable cost. (3 marks))

FIXED COST & VARIABLE COST

Fixed cost is the cost which is incurred by the firm on fixed factor. Fixed cost consists of salary of the permanent staff, rent of the building etc.

Variable cost is the cost which is incurred on variable factors. Main types of variable cost are: expenditure incurred on raw materials, wages and salaries paid to casual workers etc. Total cost is the sum of fixed and variable costs.

<u>FIXED COST</u>	<u>VARIABLE COST</u>
It is incurred on fixed factors like machines ,buildings etc	It is incurred on variable factors like labour raw materials ,transport etc.
It is independent with respect to level of output	It varies with changes in the level of output
It cannot be changed in the short run	It can be changed in the short run
It will never be zero even if there is no production	It will be zero when production is stopped
Graphically it is parallel to x-axis	It is inverse S -shaped

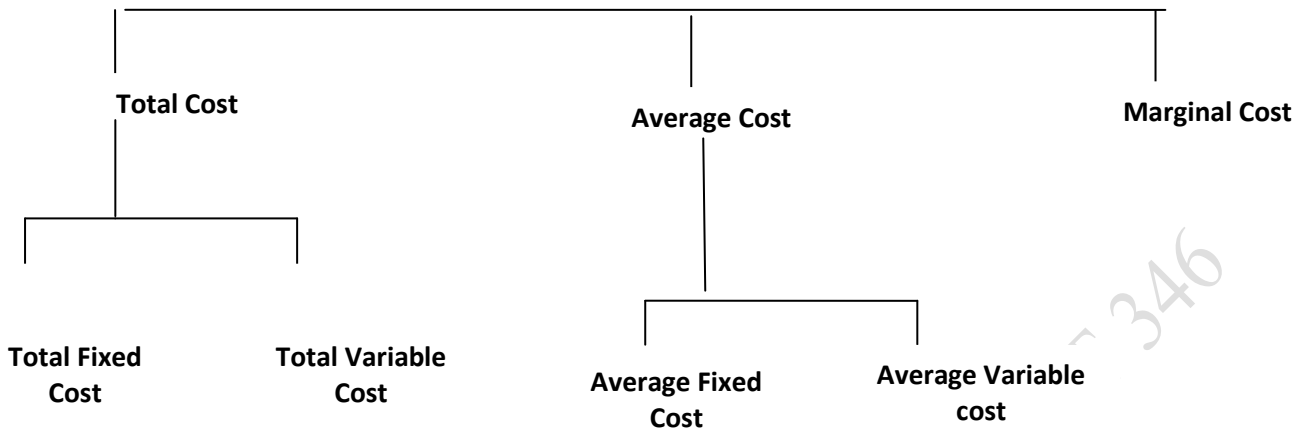
Define opportunity cost (3 marks))

OPPORTUNITY COST: The opportunity cost of any good is the next best alternative good that is given up to produce this good. **Here the cost of producing a quantity of a commodity is measured in terms of quantity of some other commodity that could have been obtained instead.** The opportunity cost arises because of the scarcity of resources and thereby arises problem of choice. For example, given resources may be used for the production of cloth or bread. If a given amount of resource can produce 1 metre of cloth or 20 loaves of bread, then the opportunity cost of 1 metre of cloth is 20 loaves of bread which need to be sacrificed in order of produce 1 metre of cloth. The concept of opportunity cost is fundamental to the basic economic problems.

BEHAVIOUR OF COST IN THE SHORT RUN

Theory of cost is primarily concerned with the behavior of cost in relation to the change in output .We can consider cost in term of total, average and marginal costs. Accordingly short run cost may be classified as follows.

SHORT RUN COST



State the relation between total cost, total fixed cost& total variable cost.(8)

Total cost: Total cost is defined as aggregate of all costs of producing any given level of output .It is the total expenditure incurred by a firm for obtaining fixed factors of production required for the producing of a commodity.

Total Cost consists of Total Fixed Cost& Total Variable Cost

Total Fixed cost: Total fixed cost refers to the total cost incurred by the firm on fixed factors. This cost is independent of output .It does not change with the change in the quantity of input. Fixed costs are fixed obligations of the firm which must be incurred by the firm whether the output is small or large. Fixed cost is often known as **unavoidable cost**. Fixed costs includes interest on the capital, rent of the building etc.

Total Variable cost: Total variable cost refers to the total cost incurred by a firm on the use of variable factors. This cost includes payments for raw materials; wages paid to temporary and casual workers payments for fuel and power used etc. These costs vary directly with change in the volume of output; rising as more is produced and falling as less is produced. That is why variable cost is also called **avoidable cost**.

**TABLE:1 TOTAL FIXED COST, TOTAL
VARIABLE COST, TOTAL COST**

OUTPUT (1) (Units)	TFC(2) (Rs)	TVC(3) (Rs)	TC(4) (Rs)
0	60	0	60
1	60	40	100
2	60	76	136
3	60	102	162
4	60	132	192
5	60	170	230
6	60	222	282

As table 1 shows total variable cost changes with change in output. It would be seen that total variable cost increases with increase in output. But the rate of increase in the total variable cost is different at different levels of output. As can be seen from the column 3 of the table, total cost increases at a decreasing rate as total output increases up to 3 units and subsequently it increases at an increasing rate with increase in output from 4th unit onwards.

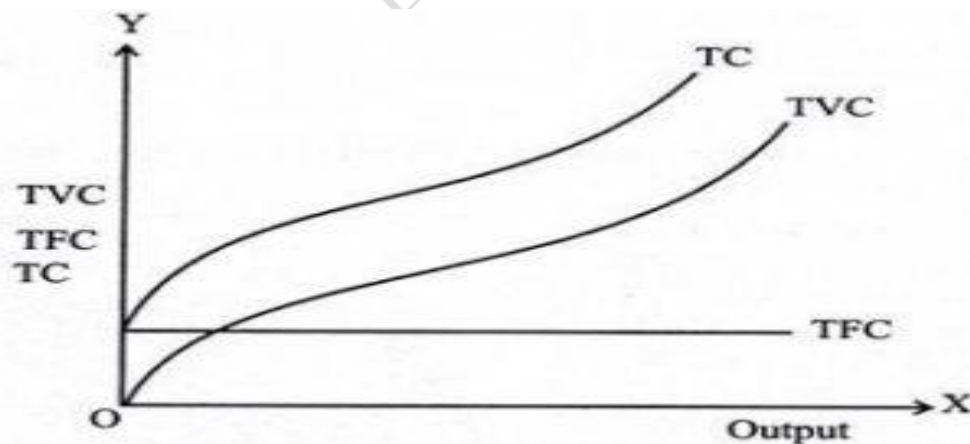


Fig: 1. Behavior of Short- run Total Costs

Figure 1 shows Total cost curve, Total fixed cost curve and Total variable cost curve. The x-axis measures output and Y-axis measures cost. The TFC curve is a straight line parallel to X –axis indicating the same amount of fixed cost at every

level of output. The TFC is constant at OA. The TFC starts from point A on the Y-axis indicating that total fixed cost will be incurred even if output is zero.

The TVC curve is **positively sloping curve showing that as output increases total variable cost also increases**. Note that TVC curve starts from the origin which shows that when output is zero, total variable cost is also zero. **The total variable cost increases at a decreasing rate and at an increasing rate subsequently with increase in total output. This follows directly from the law of variable proportions**. Higher productivity lower cost and vice versa. The shape of TVC curve is **reverse S shaped**.

The total cost curve is obtained by adding up vertically TFC curve and TVC curve because total cost is the sum of total fixed cost and total variable cost.

AVERAGE COST or AVERAGE TOTAL COST

Average cost is the cost of producing per unit of the commodity. AC is obtained by dividing total quantity by the level of output. Average cost is **U-shaped**.

$$AC = \frac{TC}{q}$$

Average cost consists of Average fixed cost and Average variable cost

$$AC = AFC + AVC$$

Average Fixed Cost: AFC is defined as the fixed cost of producing per unit of the commodity. It is obtained by dividing TFC by the level of output.

$$AFC = \frac{TFC}{q}$$

The AFC curve is derived from TFC curve is a **rectangular hyperbola**. It shows declining values of fixed cost per unit of output. It slopes downward throughout its length from left to right showing continuous fall in average fixed cost with an increase in output. For every small outputs, average fixed cost is high and for large it is low.

Average Variable Cost: AVC is the variable cost of producing per unit of the commodity. It is obtained by dividing the TVC by the level of output.

$$AVC = \frac{TVC}{q}$$

The AVC curve is derived from TVC curve and is **U-shaped**. It shows that as output increases the value of AVC falls till it reaches a minimum point. Beyond this point AVC starts rising. The reason behind the U-shape of AVC curve is the law of variable proportions.

TABLE: 2 .RELATIONSHIP BETWEEN VARIOUS SHORT- RUN COST CURVES

Output	TFC	TVC	TC	AFC	AVC	ATC	MC
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
			2+3	2÷1	3÷1	5+6	TC _n -TC _{n-1}
0	60	0	60	-	-	-	-
1	60	40	100	60	40	100	40
2	60	76	136	30	38	68	36
3	60	102	162	20	34	54	26
4	60	132	192	15	33	48	30
5	60	170	230	12	34	46	38
6	60	222	282	10	37	47	52

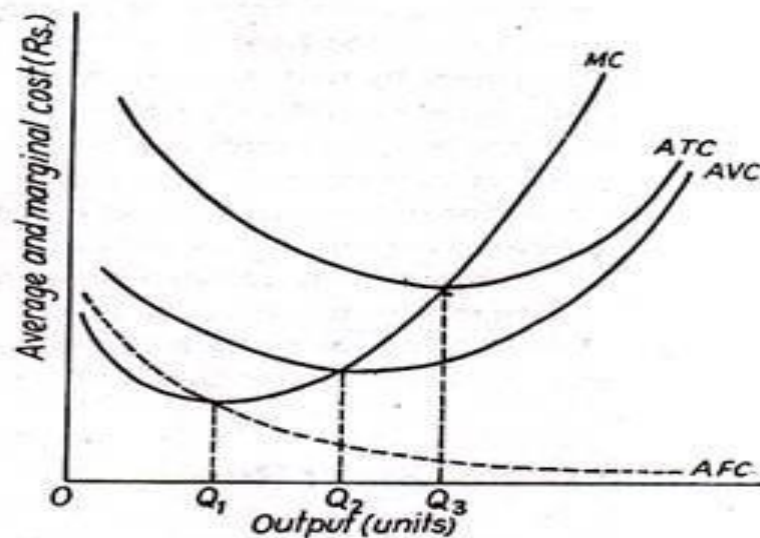


Fig: 2. Behavior of Short-run Average Cost & Marginal Cost

The ATC curve is U-shaped indicating that the average total cost falls initially, reaches the minimum point and then starts rising. **Remember that the level of output at which the average cost is minimum is known as the optimum level of production**

DEFINE MARGINAL COST. (3)

MARGINAL COST

Marginal cost is defined as the addition made to total variable cost or total cost when one unit of output is produced. It is the addition to total cost of producing n units instead of $n-1$ units where n is a given number.

$$MC_n = TC_n - TC_{n-1}$$

$$MC = \frac{\Delta TC}{Q} \quad MC = \frac{\Delta TVC}{Q}$$

In the table on page the marginal cost of 4th unit is the change in total cost when the output is increased from 3 units to 4 units. The marginal cost of the 4th unit is 30.(Rs 192-162=30).

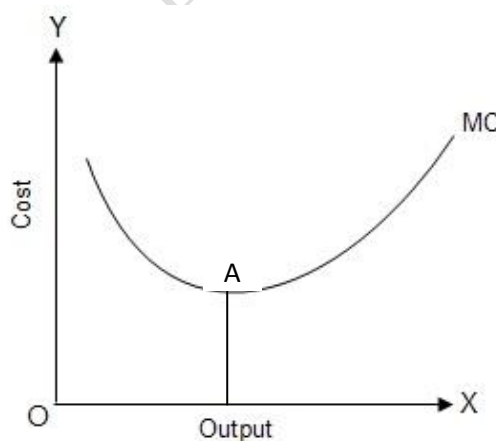


Fig:3 .Short-run Marginal Cost curves

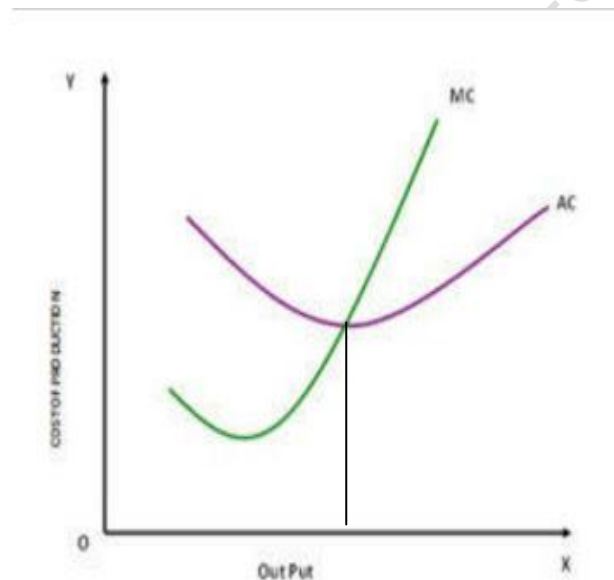
1. Marginal cost is derived from TVC curve and it is **U-shaped**. **The reason behind its shape is law of variable proportions**. As output increases MC curve slopes downward up to OQ, reaches minimum point A and then starts upward beyond OQ level of output.

2. **Marginal cost is independent of total fixed cost. It is associated with total cost and total variable cost. Marginal cost is the addition or total cost or total variable cost when the output increases. There is no change in fixed cost as output increases. MC curve is derived from TVC curve is same as derived from TC curve. Therefore MC curve is common both to AVC curve and ATC curve.**

State the relation between marginal cost & average cost. (6)

RELATION BETWEEN MARGINAL COST & AVERAGE COST

There is an important relationship between a marginal cost and average cost curves. The relationship between two is shown graphically on figure 4.



Relationship between Average Cost & Marginal Cost

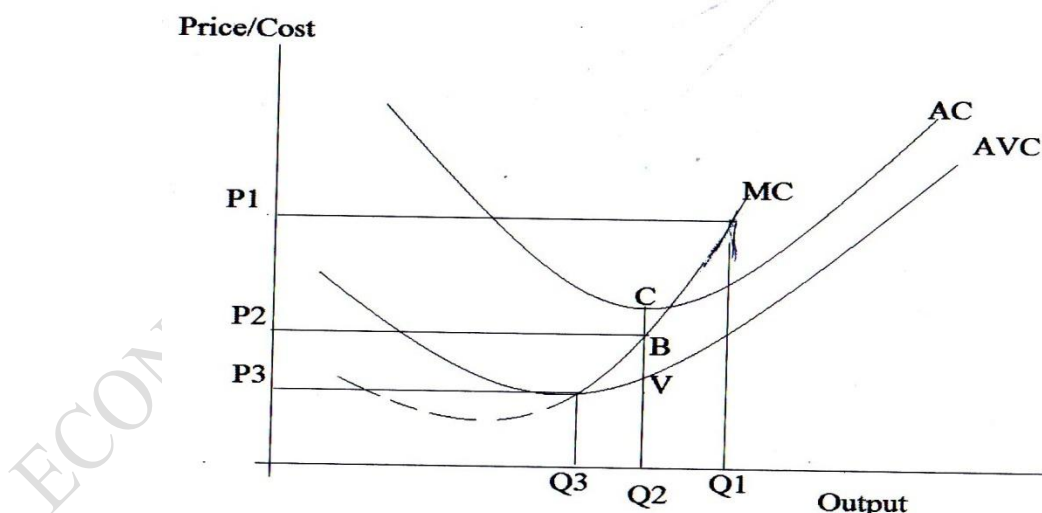
1. **When marginal cost is less than average cost, average cost falls with the increase in output.** It will be seen from the figure that as long as MC curve lies below the AC curve up to OQ amount of output, AC curve is falling.
2. **When marginal cost is greater than average cost, average cost is rising.** In the figure MC lies above AC curve beyond OQ level of output and during this range, the AC curve is rising.
3. **When marginal cost is equal to average cost, the average cost is constant.** In the figure the average cost is minimum and constant for a while at point A on AC curve. **At this minimum point marginal cost equals average cost.**

Using the above relationship between average cost and marginal cost, we can generalize that **when average cost is U-shaped, the corresponding marginal cost will be also U-shaped and it will always cut the average cost at its minimum point from below.**

SHUT DOWN POINT IN THE LONG RUN

Define shutdown point. (3 marks)

In a firm if $AVC < P < AC$, will the firm shut down or continue to produce? Give reason. Where, P =price, AVC -Average Variable Cost, AC =Average Cost.(3)A shut down point is a point **where the firm has no benefit for continuing operations.** This happens when the market price for the product is equal to the Average Variable cost in the short run. This is the minimum point of the AVC curve. At this point the firm can minimize its loss only by not producing. As long as the market price is above the AVC of the firm, in spite of making losses, the firm will cover all its variable costs and will wait until and hope to cover its fixed cost in the long run.



Shutdown point in the Short-run

In the above figure when the price is P_1 the firm will supply a quantity Q_1 where price equals marginal cost. Since this price is greater than average cost the firm is getting a profit. When price falls to P_2 , it is less than AC and hence there is a loss.-

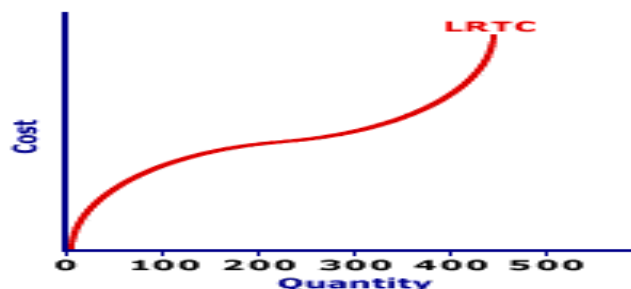
BC. But still it is beneficial for the firm to produce in the short run because when it produces Q_2 it is able to cover its variable cost Q_2V and a part of fixed cost VB . Hence the loss will be **BC** only. If it stops production loss per unit will be **VC**. Once the price reaches P_3 it may stop production because the firm is able to cover its variable cost only. Thus the supply curve of a firm is that portion of the MC curve which is over and above the AVC curve or the shutdown point

THE LONG-RUN COSTS

Long is a period which is sufficient to increase the quantities of all factors such as building, machinery, labour, raw materials etc. In the **long run all factors are variable. The long-run production function has no fixed factors and the firm has no fixed cost in the long -run. In the long-run the firm can build a new plant more suitable to layer output level.** A long run cost curve depicts the functional relationship between output and long run cost of production .**A long run cost curve is a sum of short run cost curves.**

LONGRUN RUN TOTAL COST (LTC)

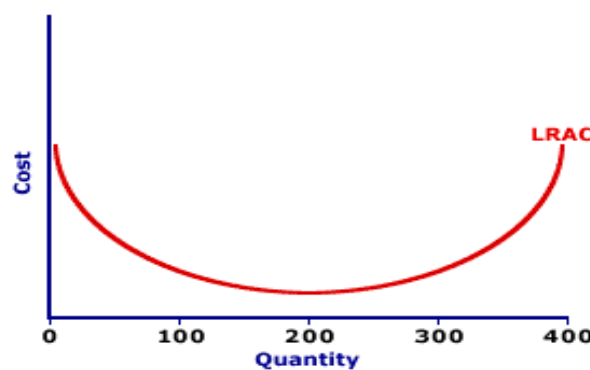
It is the minimum cost at which a given level of output can be produced in the long run.LTC is derived from the short run total cost curves .Each point on LTC is taken from a short run total cost curve corresponding to the optimum output represented by that plant size.LTC is also inverse S shape and the reason is returns to scale.



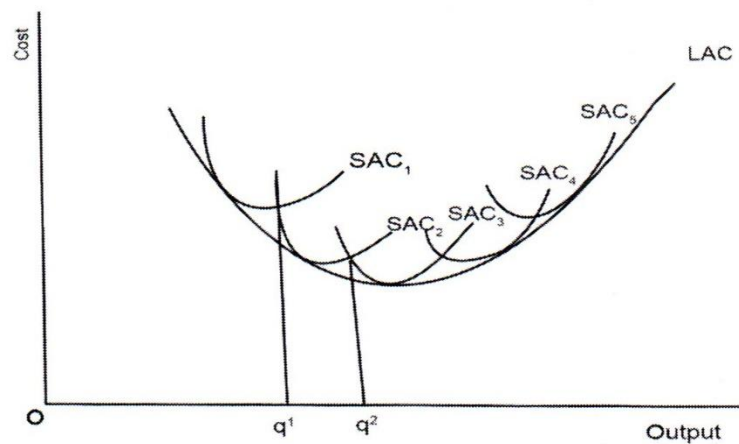
Long run Total Cost

LONG-RUN AVERAGE COST CURVE (LAC)

Long-run average cost is per unit cost of factors production in the long-run. It is obtained by dividing the long-run total cost by the level of output. It shows the lowest per unit cost of producing each level of output when all inputs have been adjusted. **Long-run average cost curve is derived from the short run average cost curves. LAC curve is derived by joining all the relevant points of SAC curves when the firm can shift from one plant to another.** The derivation of LAC is graphically depicted in the figure given below.

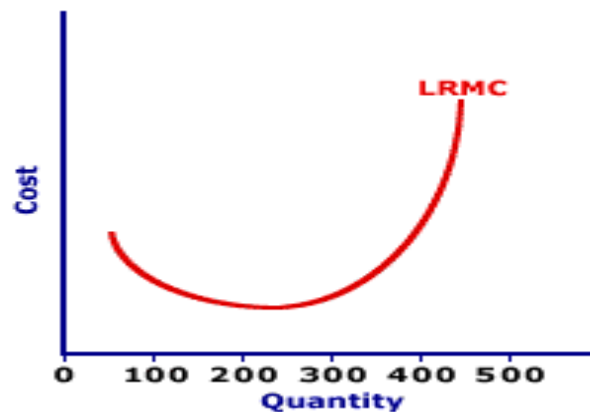


Long run Average Cost



Derivation of Long run Average Cost

LONG RUN MARGINAL COST(LMC)



Long run Marginal Cost Curve

It is the addition to total cost when one more unit of output is produced in the long run. It is derived from short run marginal cost curves. LMC curve is also U shaped because of different returns to scale in the long run.

REVENUE

Revenue is the income from the sale of output.

Total revenue (TR): It is the total receipts from the sale of a given quantity of output. It is obtained by multiplying quantity sold (Q) by price per unit (P).

$$TR = P \times Q$$

Average revenue (AR): It is the revenue per unit of output sold. AR is obtained by dividing by the number of units of output sold (Q).

$$AR = \frac{TR}{Q} = \frac{P \times Q}{Q} = P \text{ that is AR is the price}$$

Marginal revenue (MR): It is the addition to total revenue by selling one more unit of output

$$MR_Q = TR_Q - TR_{Q-1}$$

Relation between TR & MR

The following relations can be observed between TR and MR

1. When MR is positive TR increases
2. When MR is zero TR is maximum
3. When MR is negative TR decreases

ECONOMIES & DISECONOMIES OF SCALE

ECONOMIES OF SCALE

“Economies of scale are cost benefits that come with producing goods in larger numbers. It is a proportionate saving in costs gained by an increased level of production.”

Stores such as Costco and Walmart are examples of economies of scale. They utilize this principle by buying huge quantities of goods in order to receive low bulk prices, and they pass on the savings to their customers.. The economies of scale can be broadly classified as

A. INTERNAL ECONOMIES

B. EXTERNAL ECONOMIES

A. Internal Economies: Internal economies depend on the size of the firm. These advantages emerge within the firm itself as its scale of production increases. There are different forms of internal economies.

1. Labour economies: Increased production allows division of labour and it increases efficiency and productivity of workers.

2. Technical economies: As a firm expands it can use the latest technology and machinery. This increases efficiency and reduces cost of production.

3. Managerial economies: When there is large scale production, specialist managers can be appointed in different departments and this increases overall efficiency.

B.External Economies: External economies mean gains available to all the firms in an industry from the growth of that industry. That is advantages accruing to a firm due to localization of the industry. The following are the important types of external economies.

1. Economies of localisation: When number of firms in the industry is located in one place, all of them derive mutual advantages. This can be in the form of availability of skilled labour, provision of better transport facilities etc.

2.Economies of by-product: The availability of waste material in large quantity from the industry may facilitate the starting up of firms in the area which produce by-products by using these waste materials.

DISECONOMIES OF SCALE

Diseconomies of scale mean disadvantages of large production. Beyond a certain limit, diseconomies surpass internal and external economies. These diseconomies increase the average cost of production and limit the further expansion of the firm. The following are the important types of diseconomies.

1. Difficulties of management: As a firm expands problems of management arises. Beyond a limit, it will be very difficult for the manager to control the organisation.. This increases the average cost of production.

2. Difficulties of coordination: In a big organization there will be a number of departments. When the size of the organization becomes too large, proper coordination between these departments will be difficult. It will affect the overall performance.

3. Difficulties in decision making: In a large firm, before taking a decision it has to consult various departments and decision making will be delayed. Hence, a firm cannot take quick decisions and make quick changes.

QUESTIONS

1. Why should the TVC increase at decreasing rate at initial stages of production?
2. If the total cost of producing 5 units of commodity is Rs 20, and that of producing 4 units is Rs 15 what is the marginal cost?
3. State the relation between marginal cost and average cost. Why are they U-shaped?
4. a) Distinguish between fixed cost and variable cost

b) In a firm if $AVC < P < AC$, will the firm shut down or continue to produce? Give reason. Where, P =price, AVC =Average Variable Cost, AC =Average Cost.
5. Explain a) Marginal cost b) Marginal Revenue c) Sunk cost

d) Private and social cost.
6. Can there be some fixed cost in the long run? Give reasons for your answer.
7. In which cost expenditure on raw material and wages of casual laborers will be included?
8. Why short run average cost is U-shaped?
9. Explain the various short run cost curves?
10. Why does the long run average variable cost curve is called envelope curve?
11. Explain the shutdown point?
12. How cost analysis is important in business decision making?
13. The long run average cost curve is U-shaped. Why?
14. Draw total fixed cost, total variable cost and total cost curves in a single diagram. State the relation between total variable cost and total cost curves.
15. Define average and marginal revenue. With the help of a figure demonstrate the relationship between average and marginal revenue.(6)

b. Draw a diagram showing AFC, AVC and TC curves. (4)

16. Complete the following table

Units of Output	TC	TFC	TVC	MC
0	70			
1	80			
2	85			

17. Calculate the average cost and total cost with the help of the following data.

Output	Marginal cost
1	10
2	8
3	6

18. Complete the following schedule

No of units of Output	TC	TFC	TVC	MC
0	100	----- -	----- -	----- --
1	150	----- -	----- -	50
2	----- ---	----- -	----- -	40
3	----- ---	----- -	120	-----

18. a) What is explicit and implicit cost?

b) Draw a diagram and explain AFC, AVC, MC and AC.

20 a) Distinguish between TFC and TVC. Draw and TVC curves.

- b) Suppose the average cost of a product is Rs20 and average variable cost is Rs 15. If the price of the product is Rs 18, will the firm continue its production in the short run or shutdown? Give reason
- 19.a) What is MC? In short run, changes in MC depend on the changes in TVC. Why?
- b) Suppose the AC of a firm is greater than price and price is greater than AVC. Will the firm produce or shut down? Give reason.
20. Selling price of product is Rs 20, $AC=22$ and $AVC =18$. Will the firm continue its production or shutdown in the short run? Give reason
21. Explain Marginal revenue and Average revenue in Perfect competition and imperfect competition with graph.
22. a) What is shutdown point?
- b) Suppose in the short run $AVC < P < AC$. Will this firm produce or shut down? Give reason. **(KTU Model Question)**
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MODULE -2

BREAK-EVEN ANALYSIS

Define Breakeven analysis. (3 marks)

State its merits & demerits.(4 marks)

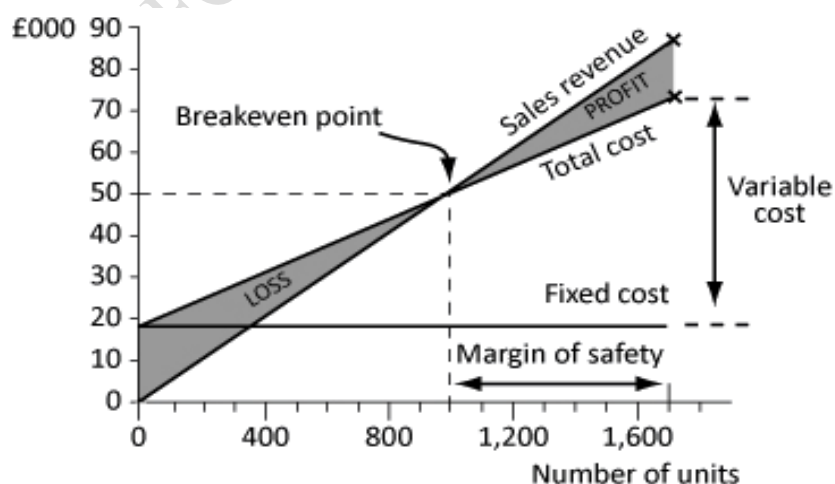
Break-even analysis is the study of relationship between cost, volume and profit at different levels of sales or production. It refers to a technique of determining that level of operations **where total revenue equals total expenses; that is the point of no profit, no loss.** The break-even point of a company is the level of sales income which will equal to the sum of its fixed and variable costs. At break-even, **sales are equal to fixed cost plus variable cost.**

$$\text{“BREAK-EVEN POINT} = \text{FIXED COST} + \text{VARIABLE COST”}$$

ASSUMPTIONS

1. Variable cost remains constant per unit of output irrespective of the level of output.
2. Fixed cost remains constant at all volumes of output.
3. There is coordination between production and sales.

BREAK-EVEN CHART



- **“Break-even chart is the graphical representation of break-even point. It shows the relationship between cost, volume of output and profit”.** It can be seen from the figure that no of units produced is shown on the X-axis and costs and sales revenue is shown along

the Y-axis. The fixed cost line is drawn parallel to X-axis which indicates that fixed expense remains the same with any volume of production. Total cost line starts with fixed cost line where variable cost line is zero. Sales line is drawn starting from the origin, which means that sales moves in relation to the level of output.

- **The sales cost line will cut the total cost line at a point where the total costs are equal to total revenue .Thus point Q is the Break-even point** where there is no profit and no loss. Thus a break-even chart helps a management to understand their business in relation to profit, requirements of capital and financial stability.

KEY DECISIONS TO BE TAKEN BY THE MANAGER

1. How many units do we need to sell to break-even?
2. What is the margin of safety?
3. How many units to be sold to achieve a targeted profit?
4. What should be the profit or loss if x quantity is sold in a month?
5. What price should be the selling price if x quantity can be sold in a month?

➤ **BREAK-EVEN POINT**

$$\text{BEP} = \frac{\text{Fixed Cost}}{\text{Selling price} - \text{Variable cost per unit}}$$

$$\text{Breakeven in sales value} = \text{Breakeven units} \times \text{Selling price}$$

➤ **MARGIN OF SAFETY**

It is the difference between actual sales and breakeven sales. It denotes the level of safety that company enjoys before company incurring losses. Higher margin of safety provides freedom to the management of the company to change the selling price of their product in order share market share from its competitors.

$$\text{MARGIN OF SAFETY} = \text{ACTUAL SALES} - \text{BREAK-EVEN SALES}$$

➤ **HOW MANY UNITS TO BE SOLD TO ACHIEVE A TARGETED PROFIT?**

$$\text{BEP} = \frac{\text{Fixed cost} + \text{Required profit}}{\text{Selling price} - \text{Variable cost per unit}}$$

CONTRIBUTION

The difference between selling price and variable cost is called Contribution.

It is the contribution which is available to meet the fixed expense of the business and provide profit for the business.

$$\text{CONTRIBUTION} = \text{SELLING PRICE} - \text{VARIABLE COST PER UNIT}$$

PROFIT VOLUME RATIO: Profit volume ratio is usually called $\frac{P}{V}$ ratio indicates the relationship between contribution and sales. It shows the rate at which profit increases with the increase in the volume of output.

$$\frac{P}{V} \text{ ratio} = \frac{\text{Contribution}}{\text{Sales}} \times 100$$

$$\frac{P}{V} \text{ ratio} = \frac{\text{Sales} - \text{Variable Cost}}{\text{Sales}}$$

P/V ratio is a useful tool available to management for solving many of its problems in profit planning. A high $\frac{P}{V}$ ratio for the business indicates that a slight increase in the volume of output without any increase in fixed cost would result higher profit. Similarly a low $\frac{P}{V}$ ratio indicates low profitability to the firm. **If we take total revenue and total variable cost to estimate the ratio it will not make any difference.**

If selling price is not given BEP can be find out by using the following formula.

$$\text{BEP} = \frac{\text{Total Fixed Cost}}{P/V \text{ Ratio}} \text{ or } \text{BEP} = \frac{\text{Total Fixed Cost} \times \text{Sales}}{\text{Sales} - \text{Variable cost}}$$

Value of sales to earn a desired profit using P/V ratio and Total Fixed Cost

$$\text{BEP} = \frac{\text{Total Fixed Cost} + \text{Desired Profit}}{\text{P/V Ratio}}$$

ALGEBRAIC METHOD

Let, P be the price of the product and Qb the break-even level of output .At the break-even point total revenue equals total cost.The difference between price and average variable cost (P-AVC) is the contribution margin per unit of output sold. It is that part of the price which contributes to cover the fixed cost of the firm.

$$\text{TR} = \text{TC} \text{ (TR} = \text{P} \times \text{Qb} \text{ and TC} = \text{TFC} + \text{TVC}$$

$$\text{Therefore } \text{P} \times \text{Qb} = \text{TFC} + \text{TVC} \text{ (TVC} = \text{AVC} \times \text{Qb)}$$

$$\text{P} \times \text{QB} = \text{TFC} + \text{AVC} \times \text{Qb}$$

$$\text{P} \times \text{Qb} - \text{AVC} \times \text{Qb} = \text{TFC}$$

$$\text{Qb} = \frac{\text{TFC}}{\text{P} - \text{AVC}}$$

	<u>Merits</u>	<u>Demerits</u>
1.	It explains the relationship between cost, volume and returns.	1. It may give misleading decisions of production capacity changes.
2.	It indicates the lowest amount of business activity to prevent losses.	2. The total fixed cost may not remain constant after a certain level of output as assumed by break-even analysis.
3.	It helps a firm in making its short –run tactical decisions	3. The assumption that variable cost per unit remains constant is unrealistic.

QUESTIONS

1.The financial details of a company are as below.

Variable cost per unit is Rs30;selling price per unit is Rs 40.Fixed expenses are Rs 1, 00,000.Calculate

➤ the break-even point

ii)margin of safety considering actual sales is 15000 units

iii) the selling price per unit ,if BEP is brought down to 8000 units.

2.From the following calculate the break-even point

Sales price	20 per unit
Variable manufacturing cost	11 per unit
Variable Selling Cost	3 per unit
Fixed factory overheads	5,40,000 per year
Fixed selling costs	2,52,000 per year

3. From the following data you are required to calculate

$\frac{P}{V}$ ratio b) Break-even sales with the help of $\frac{P}{V}$ ratio

Sales required to earn a profit of 4,50,000

Fixed expenses:90,000

Variable cost per unit:

Direct material=Rs 5, Direct Labour=Rs 2,Direct overheads=100% of Direct labour

Selling price per unit =Rs 12

4.From the following information ascertain by how much the value of sales must be increased by the company to break-even:

Sales:3,00,000,Fixed cost:1,50,000,Variable cost:2,00,000

$$\left(\text{Hint: Break-even point} = \frac{\text{Fixed cost} \times \text{Sales}}{\text{Sales} - \text{Variable Cost}} \right)$$

5. A company sells their product at Rs 520 per unit. Fixed cost is Rs 75000 and variable cost is Rs 310 per unit.

a) What is the break-even point?

b) What volume is needed to generate a profit of Rs 12,520?

6. a) Explain breakeven analysis with diagram.

b) If a company has the following details

Fixed cost = Rs 40,00,000, Variable cost per unit = Rs 100

Selling price per unit = Rs 200

Calculate i) Break-even quantity

➤ If the actual production quantity is 1,20,000, what will be the profit?

7. A company sells their product at Rs 650 per unit, the fixed cost is Rs 82,000 and variable cost is Rs 240 per unit

a) What is BEP?

b) What volume is needed to generate a profit of Rs 10,250?

8. a) Suppose a firm makes candles and every month it has to pay Rs. 3000 as rent and Rs. 3000 as interest charge. If the selling price of a candle is Rs 5 and cost per candle is Rs 2

i) Estimate the break-even level of output

ii) If the sales are 5000 candles, what will be the profit?

➤ To get a profit of Rs 15,000 how many candles are to be produced?

➤ If the sales are 5000 candles what is the margin of safety?

➤ Estimate profit volume ratio and break-even sales.

9. a) Calculate the Break-Even Quantity from the following data

Fixed Cost: Rs 25,000; Average Variable Cost: Rs 12; Selling Price Rs 17

10. a) What will be the break-even quantity if selling price increases by Rs 3?

10. b) The breakeven point helps a manager in deciding their future investment plans. Do you agree? Give reasons.

11. From the following details, calculate BEP.

Sales = Rs 1,00,000

Variable cost = Rs 60,000

Fixed Cost = Rs 15,000

12. Explain the importance of breakeven analysis in the decision making process of a firm?
13. Differentiate between break-even point and margin of safety.
14. What is Break-even analysis?
- b) Suppose a company produces batteries and its fixed cost is Rs 50,000/-. Its variable expense per battery is Rs 3/- and price of battery is Rs 8/- estimate
- Break-even output.
 - Number of batteries to be produced to get a total profit of Rs 25,000
- iii) What is the margin of safety if the planned sales is 12,000 batteries?
15. The owner of an Old-fashioned Berry Pies is thinking of adding new line of pies which require leasing new equipment for a monthly payment of Rs 6000. Variable cost would be Rs 2 per pie and retail price per pie is Rs 7.
- How many pies must be sold in order to break-even?
 - What would be the profit or loss if 1000 pies are made and sold in a month?
 - How many pies must be sold to earn a profit of Rs 4,000?
 - If 2000 can be sold and profit target is Rs 5000, what price should be charged per pie? **(KTU Model Question).**
16. A small-scale company, engaged in the production of manufacturing biodegradable carry bags has total sales of Rs. 20000. Its fixed cost is Rs. 6000, while its variable cost is at Rs. 12000. Calculate
- the P/V ratio
 - break-even point
 - Margin of safety at this level of sales
 - If it sells each bag for Rs. 5, how many bags should the company sell to break even?
 - Find out the sales required to earn a profit of Rs. 4000.
17. Suppose PV ratio is 0.2 and fixed cost is Rs 10000. What is the breakeven sales? If the price per unit is Rs 50 what is the breakeven level of output?
18. Suppose contribution per unit of output sold is Rs 10 and TFC is Rs 10000. What is the break-even output. If actual sales are Rs 1200 units

estimate margin of safety.

19. Suppose the fixed expense of a firm includes rent Rs 10000 and interest payment Rs 5000. It produces note books and sells it at a price of Rs 20. If the average variable cost per note book is Rs 10, estimate break-even level of output and the profit earned when it sells 2000 books.
20. The total sales of a manufacturing firm is Rs 20000 in this year. Its variable costs are Rs 8000 while its fixed costs are Rs 6000 for that year. Find the breakeven point of this firm.
21. a) Explain break-even analysis with the help of a diagram.
- b) Suppose the monthly fixed cost of a firm is Rs. 40000 and its monthly total variable cost is Rs. 60000.
- i. If the monthly sale is Rs. 120000 estimate contribution and break-even sales.
- ii. If the firm wants to get a monthly profit of Rs. 40000, what should be the sales?

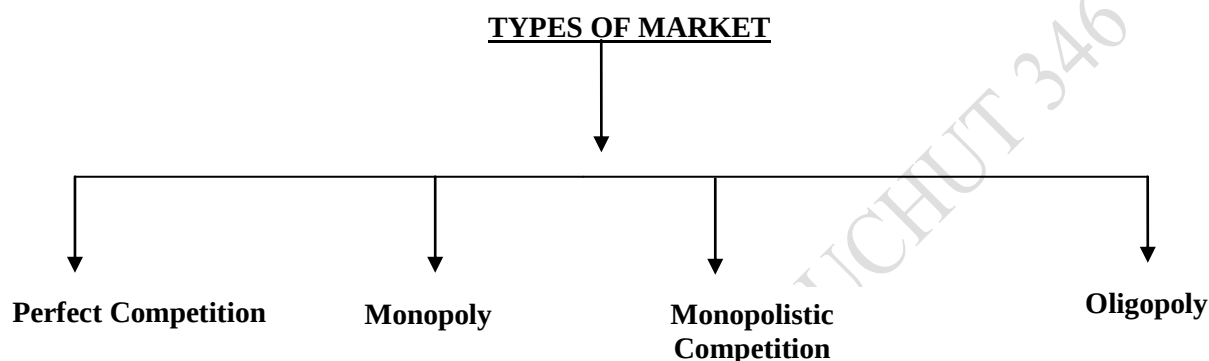
(KTU Model Question)

6. Explain break-even point **(KTU Model Question)**

22 What is margin of safety. What happens when margin of safety is low?

MODULE-3 MARKET STRUCTURES

A market is defined as place where buyers and sellers brought in close contact for the purchase and sale of a commodity. Market structures affect the supply of different commodities in the market. Economists divide market structure based on competition into four basic types.



Define perfect competition.(3 marks)

Enumerate its various features. (9 marks)

PERFECT COMPETITION

It is a market structure where an individual firm cannot influence the prevailing market price of its own. Each seller here is a price taker. Price is determined by the interaction of demand and market supply. In summary, although perfectly competitive markets are rare in the real world, some examples that closely resemble perfect competition include agricultural markets (fruits, vegetables, and grains), fish markets, stock and foreign exchange markets, online marketplaces (eBay, Etsy), and roadside flower stalls..

State the features of perfect competition. 9 marks)

1. **Large number of buyers and sellers.** The buyers and sellers are very large under this market. Each producer or seller deals with a small portion of the total output exchanged in the market.

2. **Homogenous product:** A commodity produced by all firms is same or they produce identical products. This condition ensures that the same price rules prevail in the market.
3. **Freedom of entry and exit:** There is no restriction to entry and exit. Any firm can enter the industry or they can leave the industry as they like.
4. **Profit maximization goal:** All the firms in the industry aim to maximise their profits.
5. **Free mobility of goods:** Goods can be freely transported an all parts of the market. There are no transportation cost and government regulations.
6. **Absence of transport costs:** Another condition is that there are no transportation costs in carrying the product form one place to another.
7. **Perfect knowledge regarding market conditions:** All the buyers and sellers have perfect knowledge regarding market conditions .They has clear knowledge about present and future prices and cost. Information is available of free of cost. Therefore no seller can sell the commodity the above the equilibrium price.
8. **Absence of selling costs:** under perfect competition there are no advertising costs, sales promotion etc. because all firms produce homogenous product.

FIRM & INDUSTRY

A firm is a fundamental unit which produces and sells goods and services in the economy. The purpose of production is to earn income or maximise profit.

A group of firms that produce the same or closely related products or services constitutes an industry. For example sugar industry in India covers all those firms which produce sugar in the country.

Assumptions

1. All firms produce homogenous products.
2. Cost curves of firms many vary each other.
3. All firms sell their products at the same price determined by the demand and supply of the industry so that price of each firm .

PRICE & OUTPUT DETERMINATION UNDER PERFECT COMPETITION

EQUILIBRIUM OF FIRM

Explain the price & output determination under Perfect competition using MC-MR approach? (9 marks)

MC, MR Approach: The general profit maximizing condition for a firm under perfect competition is the following

- **MC=MR.**
- **MC must cut MR from below**
- **At the point of equilibrium MC should be rising**

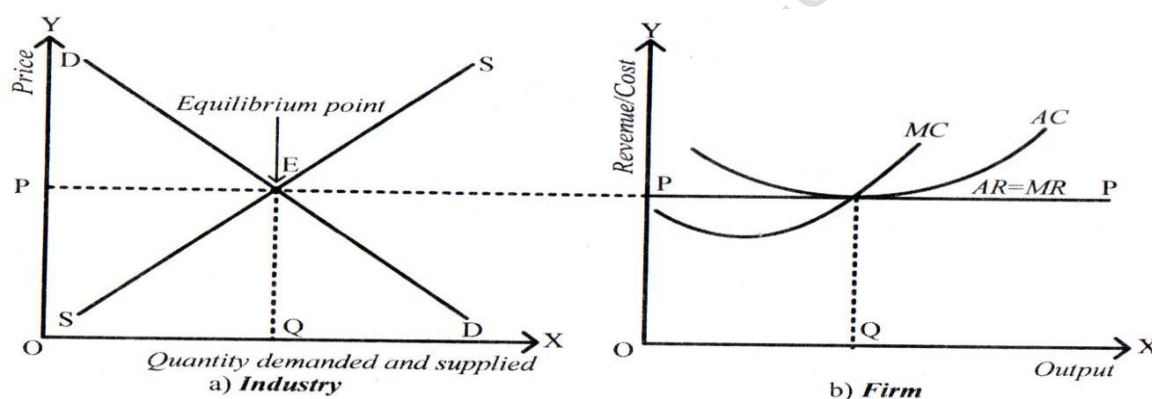


Fig: 2 Price-output determination under perfect competition MC, MR Approach

In fig 3 panel a shows the interaction of DD demands curve and SS supply at point E. Thus at equilibrium point price is OP. Panel B shows the firms demand curve. The firm will have to sell at the prevailing price OP. It may sell more units or fewer units, but can charge OP price only. The firm can neither decrease nor increase the price of the commodity because price is determined by the industry not by the firm. Firm is a price taker not a price maker. **The demand curve of the firm PP is also its average revenue and marginal revenue curve. The price is equal to average and marginal revenue because the price is constant.** Any number of commodities can be sold only at the prevailing price. At equilibrium output OQ **MC =MR. Thus equilibrium condition is $P=AR=MR=MC$**

.State the merits & demerits of Perfect competition.(3 marks)

<u>MERITS</u>	<u>DEMERITS</u>
➤ There are large number of firms so the price of the product will be low ➤ There will be efficient allocation of resources. ➤ There is no selling cost as the product is homogenous.	➤ There is little choice for consumers as the products are homogenous ➤ There is no restriction for entry of new which is not good to the industry.

QUESTIONS

1. Explain the equilibrium of perfect competition with the help of a diagram?
 2. Explain the features of perfect competition?
 3. What is equilibrium condition in a perfectly competitive market?
 4. Differentiate total revenue, marginal revenue and average revenue
 5. What are two equilibrium conditions under perfect competition?
 6. Why there is no selling cost under perfect competition?
 7. Explain the equilibrium of firm under perfect competition using TC TR approach?
 8. 'A firm is a price taker under perfect competition'. Explain
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MODULE- 3 MONOPOLY

Define monopoly. (3)

State the features of monopoly.(9)

The term monopoly refers to a firm of the product of which has no close substitutes in the market. There is a single producer for a product and he controls the entire supply of the commodity in the industry. It is very opposite of perfect competition.

Here the firm is a price- maker because it is only producer of that commodity. The public utilities like railways, post and telegraph, supply of electricity by electricity board etc. are examples of monopoly.

Describe features of monopoly? (9 marks)

1. There is a single seller but large no of buyers.
2. There will be some restrictions for the entry of new firms.
3. The producer has full control over the price and he is a price-maker.
4. There are no close substitutes for the product.
5. There is price discrimination means selling the same products at different prices to different consumers.

Explain price & output determination under monopoly using MC-MR approach?

1. MC, MR Approach: Usually a monopolist earns supernormal profit because the monopolist has complete control over the supply of a commodity. Price and output determination is explained in fig 3. The demand curve or average revenue curve is relative elastic and is downward sloping from left to right.

➤ The MR curve lies below AR curve since average revenue falls as more units of outputs are sold. **The equilibrium level of output is that level of output in which marginal revenue equals marginal cost.**

➤ The producer will continue producing as long as marginal revenue exceeds marginal cost. At the point where MR is equal to MC, the profit will be maximum and beyond this point he will stop producing.

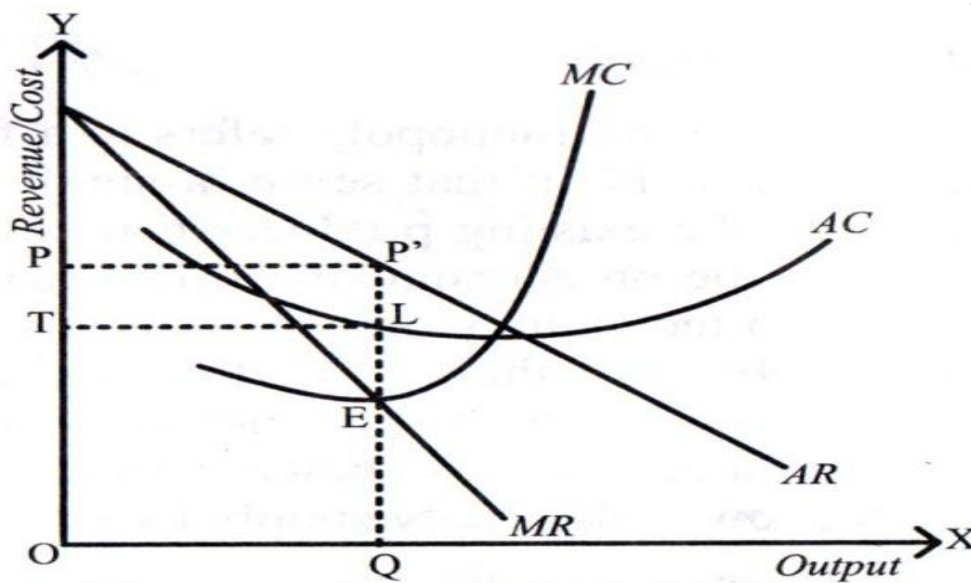


Fig: 3 Price-output determination under monopoly

It can be seen from fig3 that till OQ output, MR is greater than MC but beyond OQ the MR is less than MC. Therefore the monopolist will be in equilibrium at output OQ where MR equals to MC and profits are the greatest. The corresponding price in the figure is OP. It is clear from the diagram at output OQ, while QP is the average revenue, QL is the average cost therefore P'L is the profit per unit. The total profit is equal to P'L profit per unit multiplied by OQ the total output.

Describe the merits & demerits of monopoly?(3 marks)

Merits of Monopoly

1. They are more efficient than smaller firms as they produce at lower costs which increase in output and decrease in AVC.

2. Monopoly firms give huge amount of money which is a source of revenue for the government.

Demerits of Monopoly

1. Normally under monopoly price are higher than perfect competition.
2. A monopoly firm may exploit his labour by paying lower wages.
3. It will lead to inefficient use of scarce resources as a monopolist would produce less and charge high prices to gain more profit.

State the difference between perfect competition and monopoly.(9 marks)

DIFFERENCE BETWEEN PERFECT COMPETITION AND MONOPOLY

Basis of Difference	Perfect competition	Monopoly
No of firms	Existence of large number of firms	Existence of only one firm
Nature of product	Homogenous products are produced	There is only one firm and only single product
Price policy	There are so many firm and the seller is a price –taker	There is a single seller and he is the price - maker
Elasticity of demand	There is perfect elasticity of demand	Since there is no competition the price elasticity is less
AR&MR	The average and marginal revenue are same	The average and marginal revenue are different
Shape of demand curve	The demand curve is perfectly elastic under perfect competition because the number of firms is large and it is assumed that the product is homogenous. Therefore average revenue curve is parallel to X -axis and AR is equal to MR	AR and MR curve are different from one another .AR curves slopes downward under monopoly where MR curve slopes downward and is below AR curve.

QUESTIONS

1. What is monopoly? Explain the equilibrium of a monopoly firm?
 2. Compare the market situation of perfect competition with monopoly.
 3. A monopoly firm is in equilibrium where $MC=MR$? Explain with the help of a diagram?
 4. Explain the equilibrium of monopoly under TC, TR approach?
 5. Monopoly market will not create efficient utilization of resources. Why?
 6. What is price discrimination? Why a monopolist does practices price discrimination?
 7. Explain the merits and demerits of monopoly?
 8. Distinguish between Perfect Competition & Monopoly.
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MODULE- 3
MONOPOLISTIC COMPETITION

Define monopoly competition?

A monopolistic competition is defined as market structure in which **each seller produces a differentiated product**. There are many sellers and the product marketed by one seller can be distinguished from other seller in some form or other. Some of the important methods of product differentiation include trademarks, brand names, size, packing or colour etc.

There are close substitutes for the products under monopolistic competition. *The firm has monopoly in their product and at the same time they face competition.* It is a combination of both monopoly and perfect competition that exist in the real world.

Enumerate the features of monopoly competition?(9 marks)

FEATURES OF MONOPOLISTIC COMPETITION

1. Large number of buyers and sellers .They are selling similar products but not similar.
2. The distinct feature of monopolistic completion is product differentiation. The product of one firm differs from another in colour, shape, brand, packing brand etc. Product differentiation has many examples, in case of soaps we have brand as lux, pears lifeboy etc.
3. There is freedom of entry and exit of firms.
4. Each firm has partial control over the price of the product.
5. There is selling cost in monopolistic competition which is incurred by the producer for the promotion of his product. Advertisement expense, wages given to sales representatives etc.

Explain the price & output determination under monopoly competition? (9 marks)

PRICE AND OUTPUT DETERMINATION OF THE MONOPOLISTIC COMPETITION

In monopolistic competition profits are maximized at a point where marginal revenue is equal to marginal cost. The price determined at this point is known as equilibrium price and output produced at this point is called equilibrium output. If the marginal revenue is greater than Marginal cost the producer may plan to increase his output. On the other hand, if marginal revenue is less than marginal cost it is advisable to decrease his output. It is profitable for a producer where Marginal Revenue is equal to marginal cost. It can be seen from figure 1 that at output OQ, while OP is the average revenue. QL is the average cost, therefore P'L is the profit per unit. The total profit is equal to P'L profit per unit multiplied by OQ the total output.

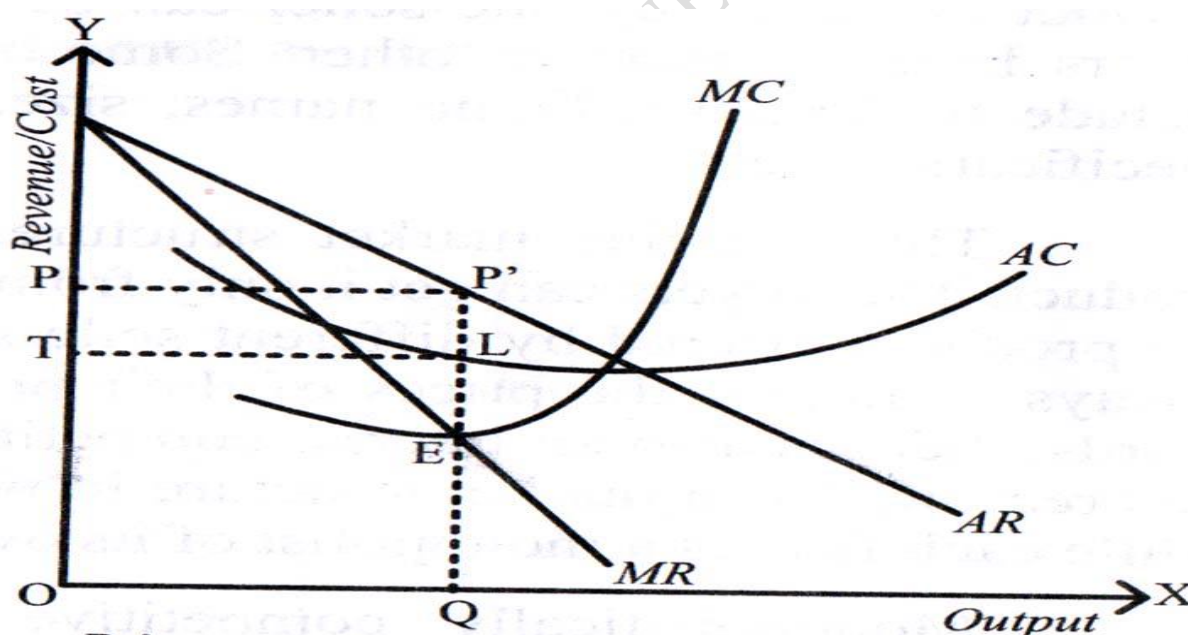


Figure 1. Price-output determination under monopolistic competition

State the merits & demerits of Monopolistic Competition.(3)

Merits of Monopolistic Competition

1. There is no restriction for entry of new firms. Therefore no single firm will find themselves with monopoly power.
2. Product differentiation will give greater choice and variety to consumers.
3. It will provide the products with high quality and service.

Demerits of Monopolistic Competition

1. Monopolistic competition is wasteful as the firms don't produce enough output efficiently to lower the average cost and benefit from economies of scale.
2. The firms enjoy some kind of monopoly power as far their product is concerned so they will sell their product at higher price.
3. The firms spend too much money on advertising their product which will indirectly lead to high price of the product and mislead the consumers.

DIFFERENCE BETWEEN PERFECT COMPETITION & MONOPOLISTIC COMPETITION

Points of Difference	Perfect Competition	Monopolistic Competition
Number of Firms	Large number of firms	Many firms but in comparison to perfect competition there are less firms.
Nature of Product	Homogenous product	Differentiated product
Price Policy	Firm is a price taker	Firm is price -maker
MR & AR	$MC=AR=Price$	$MC<AR$
Selling cost	No selling cost because the product is homogeneous	Selling cost is incurred by firms in order to increase their sales.

DIFFERNECE BETWEEN MONOPOLY & MONOPOLISTIC COMPETITION

Points of Difference	Monopoly	Monopolistic Competition
Number of Sellers	Single Seller	Large number of sellers
Nature of the product	No product differentiation because there is only one single seller	Product differentiation is a necessary the firm has close substitutes and face strong competition from rivals
Degree of Elasticity	There is less elasticity as there is no close substitutes	High degree of elasticity as there are very close substitutes
Price Discrimination	Price discrimination is possible as there is only one seller for a product	Price discrimination cannot be practised by the seller.
Selling Cost	A very small amount is spend in order to inform the consumers about the product	Huge amount is spend on advertisement in order to increase the sales of the product since there is very close substitutes.

QUESTIONS

1. Explain the features of monopolistic competition? How far it is different from monopoly?
2. Differentiate monopolistic competition and oligopoly.
3. Make a comparison between perfect competition and monopolistic competition.
4. 'A firm under monopolistic competition will only get normal profit in the long-run'. Substantiate the statement with illustrations.

5. Explain the equilibrium condition of monopolistic competitive market with the help of a diagram?
 6. What is product differentiation?
 7. The elasticity under monopolistic competition is higher than that of monopoly. Explain?
 8. Bring out the differences in monopoly, monopolistic competition and perfect competition.
 9. Explain any four features of monopoly? Make a comparison between the demand curves under monopoly and monopolistic competition.
 10. What is predatory pricing?
 - b) Explain the equilibrium of a firm earning supernormal profit under monopolistic competition. **(Model Question)**
 12. With the help of a diagram, explain equilibrium under monopolistic Competition.
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MODULE-2

OLIGOPOLY

Define Oligopoly.(3 marks)

Enumerate the features of Oligopoly. (9 marks)

Oligopoly market structure is characterized by a few sellers. The degree of competition in oligopoly structure is less than monopolistic competition, but higher than that in monopoly which has almost no competition.

The main features of an oligopoly market situation are as follows.

1. There a large number of buyers
2. There are only a few sellers
3. There are entry and exit barriers
4. The product may be homogenous or heterogeneous
5. The price and output decision of one firm are highly dependent on those of others.

In an oligopoly market an individual firm's reaction will cause its competitors to react. Such interdependence is the major characteristic of an oligopoly market. Any reaction of a firm leads to a series of actions and reactions by other firms. Thus before taking any decision for the revision for the price of its product, a firm must carefully consider the potential reactions of its competitors. Pricing in an oligopoly has been explained on the basis of several models. Few important are discussed below.

PRICE & OUTPUT DETERMINATION

Explain the kinked demand curve model of oligopoly?

What happens if a firm cuts price in oligopoly?

KINKED DEMAND CURVE MODEL

The Kinked Demand curve model was developed by Paul M Sweezy in 1939. It explains price rigidity under oligopoly on the basis of following assumptions.

- IF A FIRM INCREASES ITS PRICE OTHERS WILL NOT FOLLOW.
- IF A FIRM DECREASES ITS PRICE OTHERS WILL ALSO DO THE SAME.

Usually an oligopoly firm will not enter into a price war and hence price remains rigid. If one firm decreases the price the others will reduce the price. Hence firms demand will not increase but at the same time it will affect their profitability. On the other hand if the firm increases the price, others will not increase the price and hence it will lose its customers. Therefore a firm in oligopoly **sticks to its price**. This kind of behaviour in oligopoly was explained by the kinked demand curve. The lower part of the demand curve is less elastic because a firm cannot gain from a price cut. The upper part of the demand curve is more elastic because there will be substantial fall in demand if there is a price hike. The demand curve in an oligopoly is indeterminate. This is shown on fig1.

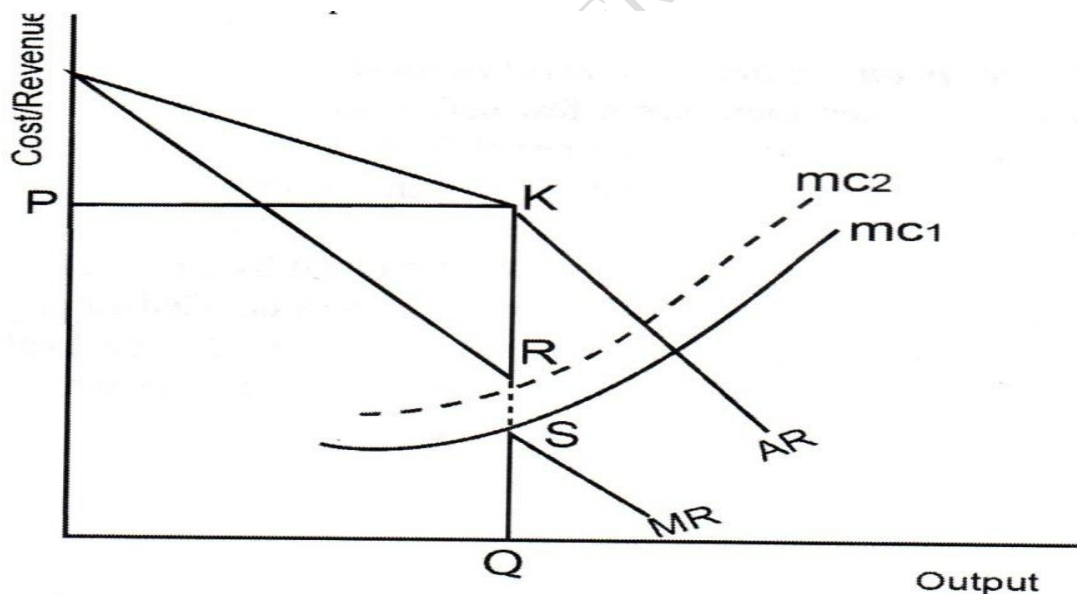


Fig 1.KINKED DEMAND CURVE MODEL

It can be seen from the figure that there is a kink at point K in the demand curve. This kink in the demand curve or AR curve at point K creates a discontinuity in the MR curve. At the kink MR remains unchanged between points R and S. Marginal cost curve mc_1 intersects at point S and OQ is the equilibrium level of output. Suppose cost increases and MC shifts upwards as mc_2 there will not be any change in equilibrium price and quantity till MC reaches the point R in the gap.

PERFECT COMPETITION	MONOPOLY	MONOPOLISTIC COMPETITION	OLIGOPOLY
Large number of buyers and sellers	Single seller and large no of buyers	Fairly large number of buyers and sellers	Few sellers and large no of buyers
Homogenous product	Single product	Differentiated product	Homogenous or Differentiated product
Free entry and exit	Closed entry	Free entry and exit	Barriers to entry
Price-taker	Price maker	Price maker	Price maker
The demand curve perfectly elastic	Negatively sloped less elastic demand curve	Downward sloping more elastic demand curve	Indeterminate demand curve
Profits are normal	Supernormal profits	Profits are normal	Supernormal profits
No selling cost	No selling cost	There is selling cost	There is selling cost
$AR=MR$	$AR>MR$	$AR>MR$	$AR>MR$
No market power	Significant power	Slight market power	Large market power

QUESTIONS

1. Explain the price and equilibrium under oligopoly market?
2. Explain the features of oligopoly market?
3. Differentiate monopolistic competition and oligopoly.
4. What will happen if a firm cut price in an oligopoly market? Why price in oligopoly remains rigid?

5. The demand curve of an oligopolistic firm is indeterminate. Why?
 6. Explain the kink demand curve model of oligopoly?
 7. With a suitable diagram explain equilibrium price and determination of a firm under oligopoly?
 8. What is a kinked demand curve?
 9. Explain the important features of oligopoly
 10. Distinguish between monopoly & oligopoly.
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